

QIP 2020

Program Committee Report

Andrew Childs
QuICS, University of Maryland

Role of the PC

QIP Charter: The role of the Program Committee (PC) is to select the best submitted papers and to put together a programme for the next QIP. The PC chair chooses the members of the PC (helped by advice from the SC), typically 15 or more people representing the broad range of subfields, including both computer science and physics. The PC chair determines (in cooperation with the SC) the rules for submission and puts out a call for submissions. The selection of talks among the submitted papers is competitive, with typically between 40 and 60 accepted submissions; it is recommended to keep the number of acceptances minimal but the PC has flexibility to accommodate exceptional breadth and quality of submissions in a given year. The PC can designate a few of the best submissions as “plenary” talks (given by “plenary speakers”), which are assigned more time in the program than regular contributed talks (given by “regular speakers”). All regular contributed talks are given in time slots ranging from 20-40 minutes.

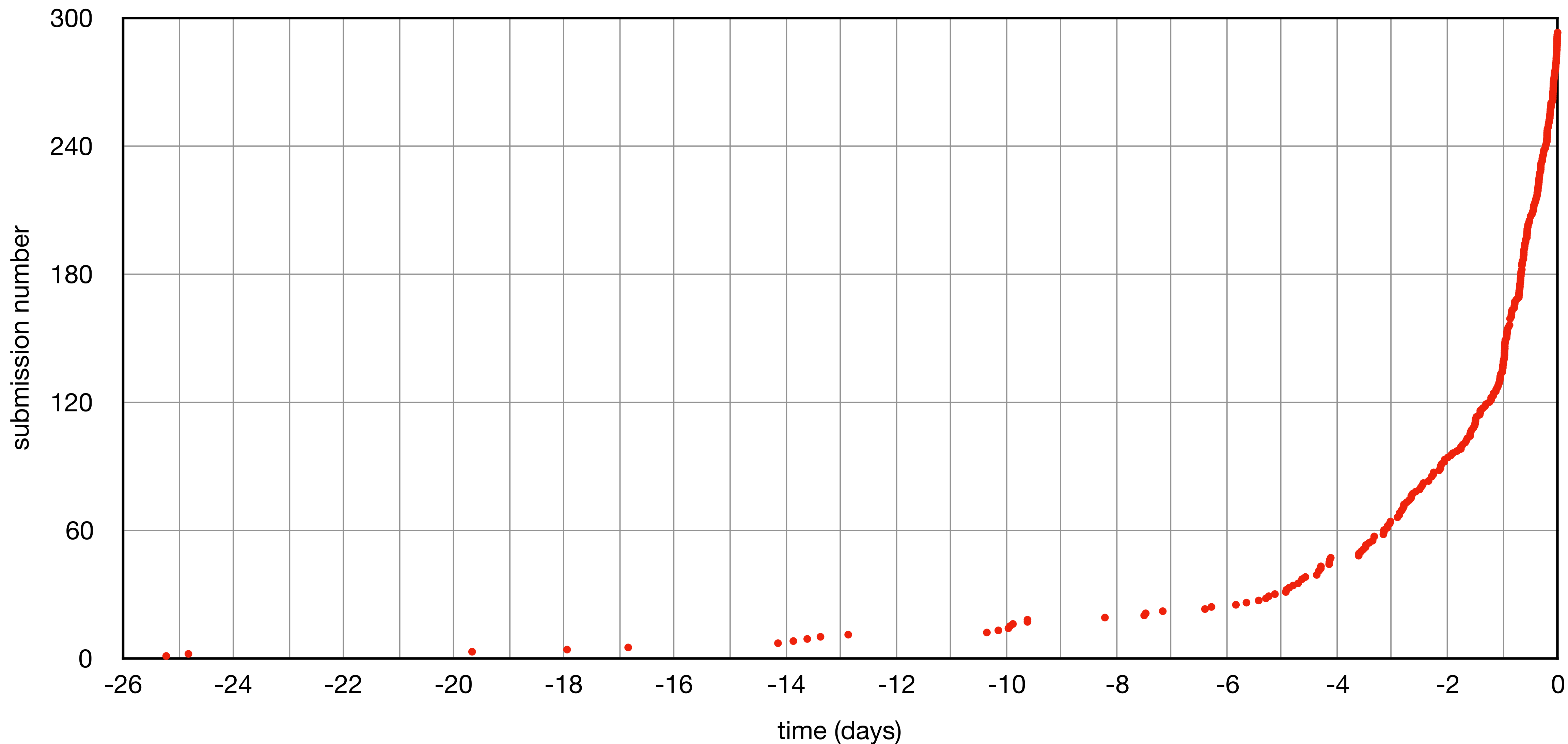
Poster submissions will generally be accepted unless they are off-topic or clearly wrong, to enable people to obtain funds for travel. At the discretion of the PC, best poster prizes can be awarded.

Program Committee

Michael Ben-Or (Hebrew University)
Aleksandrs Belovs (University of Latvia)
Mario Berta (Imperial College London)
Adam Bouland (UC Berkeley)
Michael Bremner (University of Technology Sydney)
Daniel Brod (Universidade Federal Fluminense)
Earl Campbell (Sheffield University)
Andrew Childs (University of Maryland)
Giulio Chiribella (The University of Hong Kong)
Andrea Coladangelo (Caltech)
Matthew Coudron (University of Waterloo)
Sarah Croke (University of Glasgow)
Elizabeth Crosson (University of New Mexico)
Runyao Duan (Baidu/University of Technology Sydney)
Omar Fawzi (ENS de Lyon)
David Gosset (University of Waterloo)
David Gross (University of Cologne)
Aram Harrow (Massachusetts Institute of Technology)

Zhengfeng Ji (University of Technology Sydney)
Elham Kashefi (University of Edinburgh/CNRS, Sorbonne U.)
Mária Kieferová (University of Technology Sydney)
Robin Kothari (Microsoft)
Christian Majenz (Centrum Wiskunde & Informatica)
Anand Natarajan (Caltech)
Miguel Navascués (Austrian Academy of Sciences)
Jonathan Oppenheim (University College London)
Ana Belén Sainz (ICTQT, University of Gdansk)
Florian Speelman (Centrum Wiskunde & Informatica)
Aarthi Sundaram (University of Maryland)
Ewin Tang (University of Washington)
John Watrous (University of Waterloo)
Nathan Wiebe (Pacific Northwest National Laboratory)
Mark M. Wilde (Louisiana State University)
Penghui Yao (Nanjing University)
Henry Yuen (University of Toronto)

Submissions over time



Review process

- Submissions, reviewing, and notifications handled through EasyChair.
- PC members declare conflicts of interest. PC chair's conflicts handled by the Vice Chair (John Watrous, PC chair of QIP 2013).
- Every paper was reviewed at least 3 times. There were 857 total reviews. A typical PC member handled 25 or 26 papers.
- PC members could solicit subreviews, but were responsible for editing reviews and determining scores consistent with all their assigned papers.
- PC spent 2.5 weeks discussing papers (1409 comments). Selection based on rounds of discussion to reach consensus and phases of championing and voting on papers. Also used nominations and voting to select plenary, single-track, and best student talks.
- The review process is imperfect and subjective. We did our best to be fair and to select the best possible program in the available time.

Statistics

- **283** talk/talk-or-poster submissions (cf. 349 at QIP 2019, 296 at QIP 2018)
- 78 accepted submissions
 - 5 merged in pairs
 - **73** talks selected for the program by PC (not including 1 talk selected as plenary that was independently selected by the SC)
- $78/283 = \mathbf{28\%}$ acceptance rate (cf. 18% at QIP 2019, 21% at QIP 2018)
- 2 talks promoted to plenary
17 talks promoted to single-track
- 2 talks withdrawn after acceptance
 - **71** PC-selected talks in the final program (cf. 60 at QIP 2019, 57 at QIP 2018)
- About **350** poster submissions (all accepted), including talk-or-poster submissions that were not accepted as talks

Comments

- The PC must make hard choices about what talks to include. Lots of good work is not accepted.
- Nevertheless, accepting significantly more talks would significantly reduce average quality.
- A longer review phase might help (to get more confidence in correctness of results and to make decisions on borderline papers less arbitrary). This would require starting the entire process much earlier.
- The QIP charter currently requires a student to present the talk for a submission to be eligible for the best student paper prize. The SC should consider revising this to make exceptions for visa issues.